Concentration (M)	Molar conductance ($\times 10^{-3} Sm^2\ mol^{-1}$)		
	NaCl	KCl	HCl
0.1	10.674	12.896	39.132
0.01	11.851	14.127	41.20
0.0001	12.374	14.695	42.136

Based on the above such results, Kohlraush deduced the following empirical relationship between the molar conductance (Λ_m) and the concentration of the electrolyte (C).

$$\Lambda_m = \Lambda_m^\circ - k\sqrt{C} \qquad .....(9.11)$$

The above equation represents a straight line of the form $y = mx + c$. Hence, the plot of Λ_m Vs $\sqrt{C}$ gives a straight line with a negative slope of –k and the y intercept, Λ_m°. Where Λ_m° is called the limiting molar conductivity. i.e., the molar conductance approaches a limiting value in very dilute solutions.

For strong electrolytes such as KCl, NaCl etc., the plot, Λ_m Vs $\sqrt{C}$, gives a straight line as shown in the graph (9.4). It is also observed that the plot is not a linear one for weak electrolytes.

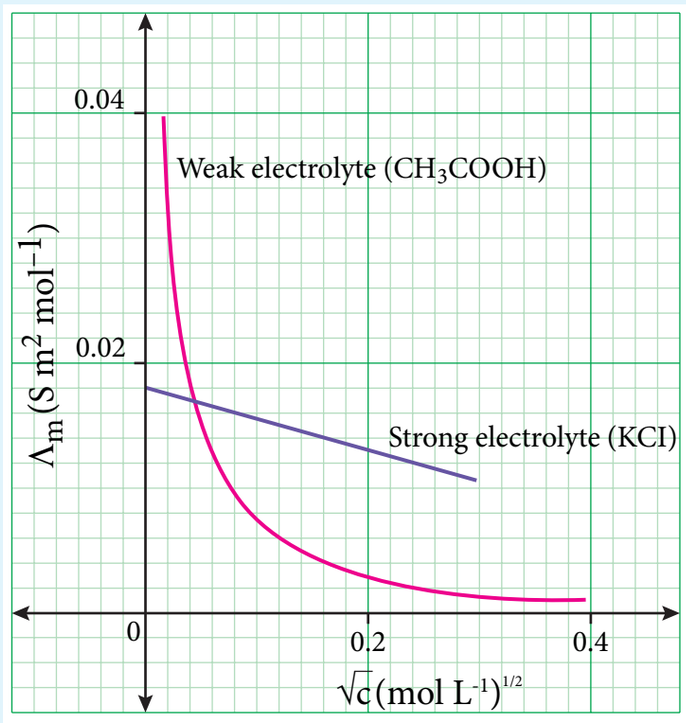


Figure 9.4 Variation of molar conductance with concentration

For a strong electrolyte, at high concentration, the number of constituent ions of the electrolyte in a given volume is high and hence the attractive force between the oppositely charged ions is also high. Moreover the ions also experience a viscous drag due to greater solvation. These factors attribute for the low molar conductivity at high concentration. When the dilution increases, the ions are far apart and the attractive forces decrease. At infinite dilution the ions are so far apart, the interaction between them becomes insignificant and hence, the molar conductivity increases and reaches a maximum value at infinite dilution.

For a weak electrolyte, at high concentration, the plot is almost parallel to concentration axis with slight increase in conductivity as the dilution increases. When the concentration

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